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A Newly Synthesized Hydrocarbon Based Chemical Demulsifier for the Treatment of Oil-Water Emulsion

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Abstract

This study investigated the demulsification efficiency (%DE) of a laboratory grown hydrocarbon based non-ionic demulsifier (ND) to disrupt stable crude oil emulsions into oil and water. Two different emulsions were formulated from oil to water ratios of 5:5 and 3:7, ND dosages of 50, 100, 150 and 200 ppm were injected in the formulated emulsions to monitor oil/water separation examined at demulsification temperature of 85 °C using the bottle test method. Demulsification mechanism and destabilization kinetics of ND was assessed via bulk emulsion viscosity and Turbiscan determinations at 25 °C and 70 °C, respectively. The outcome of demulsification activities for emulsions derived from oil to water ratios of 5:5 and 3:7 revealed that ND caused oil/water separation across all the tested dosages with optimal %DE of 96% and 99.86% achieved at 100 ppm ND for the examined oil to water ratios. Viscosity determination as a function of time showed that the incorporation of 100 ppm ND in the formed emulsion replaced the emulsifying materials around the emulsion droplets which led to reduction in emulsion viscosity as compared to the viscosity of emulsion without ND. Destabilization kinetics of the produced emulsions analyzed by Turbiscan showed that emulsion without ND absorbed most of the transmitted light without any noticeable oil/water separation while increased passage of light signals across the entire scan were perceived in the emulsion filled with 100 ppm ND indicating light transmission through the separated water. It is envisaged that this new chemical demulsifier would be of immense benefit in the petroleum and petrochemical sectors.

Keywords: non-ionic, viscosity, kinetics, demulsification, emulsion, demulsifier

Introduction

As oilfields become mature, their production rate diminishes significantly because of the attainment of optimum productivity and they are close to economic limit (Ding et al. 2020). Owing to the reservoir depletion, crude oil produced is comingled with more water volume giving rise to emulsified crude oil. These two immiscible fluids become stable emulsion not only by the presence of asphaltenes in crude oil, but also due to the concurrent action of shear and pressure drop at wellhead, valves and chokes (Raya et al. 2020). All forms of emulsions can be encountered in oilfields; however, water-in-oil emulsion is more

prevalent. In this type of emulsion, small water droplets are dispersed within an external (continuous) oil phase and it accounts for about 95% of crude oil emulsion generated in oilfields (Wong, Lim, and Dol 2015). The formation occurs once the mixtures flowing in the piping valves and porous rocks are able to withstand the turbulence at high temperature-high pressure conditions. As a result, the emulsion formation is further enhanced because of the presence of surface-active agents at oil-water interface, water pH, and ionic compositions. Highly stable crude oil emulsions are therefore a great concern for operators in the upstream and midstream processes and efficient breaking or separation of oil and water becomes paramount before crude oil transportation by pipelines to the refinery. Demulsification is the term used to describe the method or mechanism of breaking crude oil emulsion into oil and water phases. Different demulsification strategies such as microwaves (Fortuny et al. 2007a, 2007b; Guzmán-Lucero et al. 2010; Lemos et al. 2010), electrical (Han et al. 2018; Kokal 2005; Kokal and Aramco 2005; Larson, Raghuraman, and Wiencek 1994; Yonguep et al. 2022a; Zolfaghari et al. 2016), mechanical (Peng et al. 2018a, 2018b; Yonguep et al. 2022b, 2022a) and chemical (Abdullah et al. 2022; Abdullah and Al-Lohedan 2020; Husain et al. 2021; Wei et al. 2021) have been deployed by researchers to split emulsions and reduce water present in the crude oil. Among these techniques, chemical demulsification is currently gaining considerable attention because there is no need for the deployment of heavy equipment before it can be applied and a little amount of an active and efficient chemical demulsifier can destabilize huge volume of emulsion leading to effective water removal. Polymeric surfactants (Pensini et al. 2014; Shehzad et al. 2017), ionic liquids (Abdullah et al. 2022; Atta, Al-Lohedan, and Abdullah 2016; Bera et al. 2020; Husain et al. 2021), magnetic nanoparticles (Duan et al. 2017; Wang et al. 2018; Xu et al. 2018) and non-magnetic nanoparticles (Javadian and Sadrpoor 2019, 2020; Kumar, Raman, and Aichele 2018) are the notable chemical demulsifiers that have been reported in the literature while ionic liquids remain the widely used chemical demulsifiers in oilfields owing to its unique characteristics such as tunability, low volatility and the ability to dissolve in many organic and inorganic solvents. Despite these excellent features of ionic liquids, petroleum operators in the midstream are searching for new chemicals that would perform like ionic liquids or even better. This is because ionic liquids are very expensive to produce and the synthesis route is not simple. Likewise, the toxicity and non-biodegradable of many ionic liquids are of great environmental concern, hence, producing new chemical demulsifiers with high efficiency for oil and water separation becomes essential in order to address the challenges of huge cost implications and toxicity concern. The current study is therefore reporting a newly synthesized hydrocarbon based chemical demulsifier (non-ionic demulsifier [ND]) that is very cheap, less toxic and can compete favorably with commercial chemical demulsifier. Crude oil emulsions of different oil/water ratios were formulated and the performance of ND was assessed at different dosages via the bottle test technique. Rheological determination and Turbiscan analysis were employed to gain insight into the demulsification mechanism of ND.

Materials

The crude oil used in this study was obtained from one of the oilfields in Saudi Arabia and it is applied for emulsion formation without any further treatment. From the SARA analysis, this crude oil constitutes: saturate (36.2 wt%), asphaltenes (2.785 wt%), resins (11 wt%) and aromatics (50 wt%). Seawater of different salts (NaCl, Na₂SO₄, CaCl₂, MgCl₂ and NaHCO₃) having a total salinity of 57700 mg/L was used during emulsion preparation. This specific seawater concentration was used because of high salinity in the crude oil during processing. The starting chemicals and reagents for the synthesis of the developed non-ionic demulsifier (ND) were purchased from Sigma Aldrich.

Preparation of emulsion

Two sets of emulsions were formed in this study. The first batch of emulsion prepared contained equal volume (5mL) of crude oil and seawater while the second batch composed of 3 mL crude oil and 7 mL

seawater. These emulsions were formed by gradual addition of seawater in a beaker containing the crude oil and mixing was done under ambient condition for 15 minutes at a speed of 750 rpm. Prior to the demulsification test, the formulated emulsions were examined for five weeks without any noticeable oil/water separation indicating that these emulsions were stable.

Demulsification test

Demulsification efficiency (% DE) of the synthesized ND was examined via the bottle test method, and ND dosage of 50, 100, 150 and 200 ppm were injected in the prepared emulsions while reference emulsion samples (without ND) were also subjected to demulsification activity. All samples were agitated by hands for 10 seconds and demulsification performance of ND was carried out at 85 °C; oil and water separation were monitored for 1 hour. Following this, % DE was calculated using the formula (Adewunmi et al. 2021) shown in equation 1:

$$\text{Demulsification efficiency (\% DE)} = \left(\frac{V}{V_0} \right) \times 100 \quad (1)$$

Where V is the volume of separated water and V_0 represents the initial total water volume in the prepared emulsion.

Characterizations

Bulk rheology determination was employed to assess the effectiveness of ND to destabilize the emulsifying substances responsible for emulsion stability. Disruption of emulsion droplets was observed using the discovery hybrid rheometer. Emulsion viscosity measurements was done under a fixed shear rate of 0.01 s^{-1} and viscoelastic determinations were carried out at a 5% strain. The concentric cylinder geometry was used during bulk rheology tests and all tests were performed at ambient temperature (25 °C). Emulsion instability in the presence of ND was further analyzed by the Turbiscan equipment which is individually controlled and the temperature was set at 70 °C. Emulsion samples of at least 23 mL with and without ND was transferred inside the cell and scanning was performed for 24 hours. Stability analysis was carried out using the light transmission profiles through the bottom of the cell until its top. The profiles of light transmission intensity during scanning were applied to determine how stable is the formed emulsion in the presence and absence of ND.

Results and Discussion

Demulsification efficiency of ND

The synthesis pathway of lab grown non-ionic demulsifier (ND) used in this study is displayed in Fig. 1, while the details of its production can be found elsewhere (Adewunmi, Kamal, and Hussain 2024). Fig. 2(a) shows the demulsification efficiency (%DE) of 50, 100, 150 and 200 ppm dosages of ND at demulsification temperature of 85 °C for emulsion formulated from oil to water ratio of 5:5. As reflected in Fig. 2(a), emulsion without ND (blank) did not experience any noticeable oil-water separation during the entire 1 hour of demulsification activity. However, the incorporation of different dosages of ND in the formed emulsion caused the breaking of emulsion droplets and gave rise to oil and water separation. Across all the tested concentration of ND, oil-water separation proceeded at a faster rate until around 0.8 hour when the separation was nearly constant until demulsification activity reached 1 hour duration. The % DE of ND at 50, 100, 150 and 200 ppm after 1 hour was determined to be 80%, 96%, 90% and 89.8%; respectively, with 100 ppm ND exhibiting the optimal oil-water separation for emulsion produced from oil to water ratio of 5:5. In the case of emulsion generated from oil to water ratio of 3:7, oil-water separation performance of ND was seen to have increased within 1 hour of demulsification exercise and at the same temperature while the

blank sample of this specific emulsion also failed to exhibit any noticeable oil-water separation throughout the entire demulsification duration. Accordingly, after 1 hour when there was no further observable oil-water separation, the %DE of ND at 50, 100, 150 and 200 ppm for emulsion formulated from oil to water ratio of 3:7 was estimated to be 99.28%, 99.86%, 90%, and 89.29%. Again, 100 ppm ND achieved the best oil-water separation in the emulsion resulting from water ratio of 3:7. The outcomes of oil-water separation demonstrated that oil-water volume ratios in the produced emulsions had influence on the performance of ND. Oil and water separation for the emulsions generated from oil to water ratios of 5:5 and 3:7 were shown in Fig. 3 (a & b).

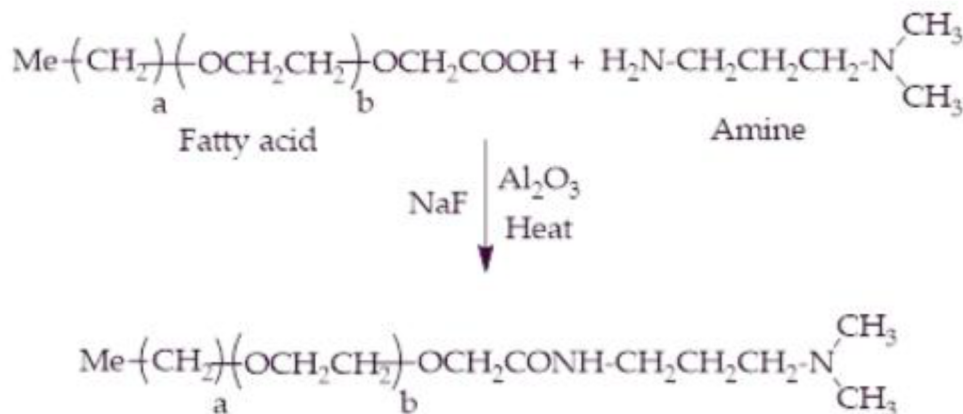
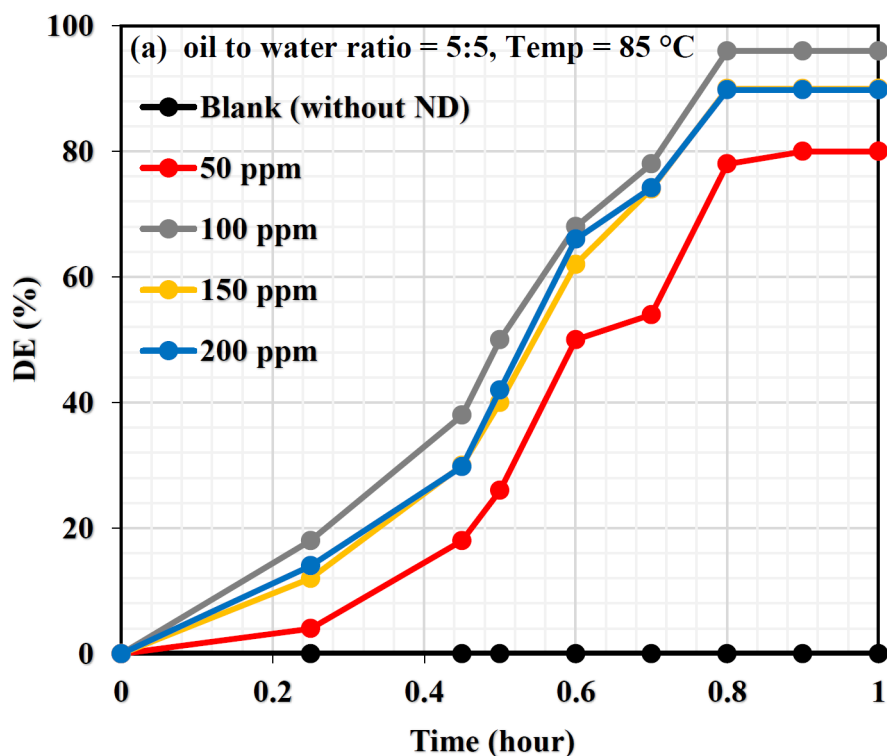


Figure 1—Synthesis pathway of the developed non-ionic demulsifier (ND).



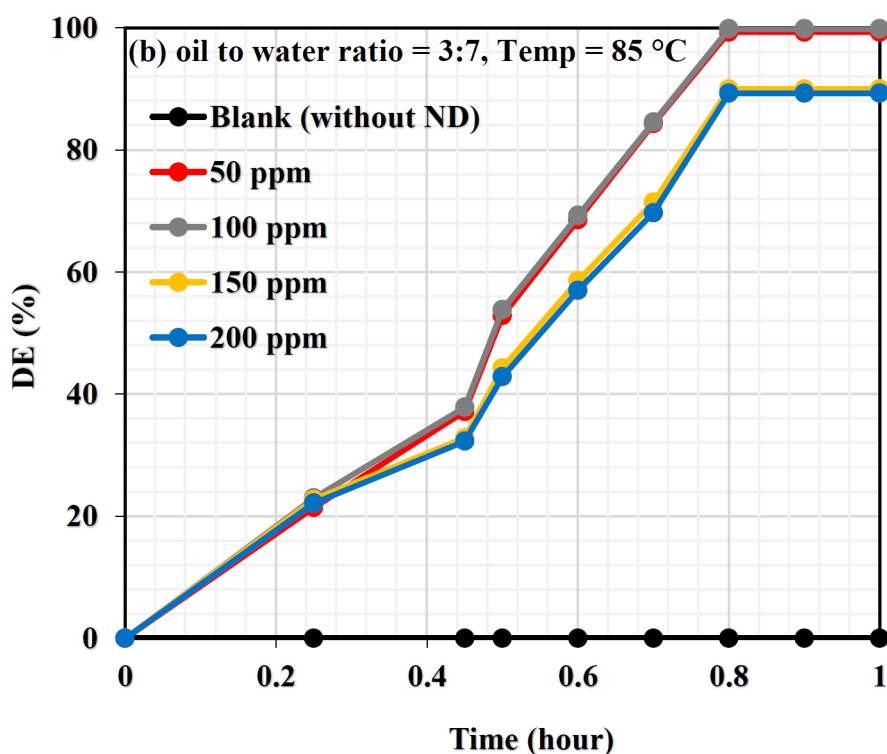


Figure 2—Demulsification efficiency (%DE) of ND at 85 °C (a) oil to water ratio is 5:5 (b) oil to water ratio is 3:7.

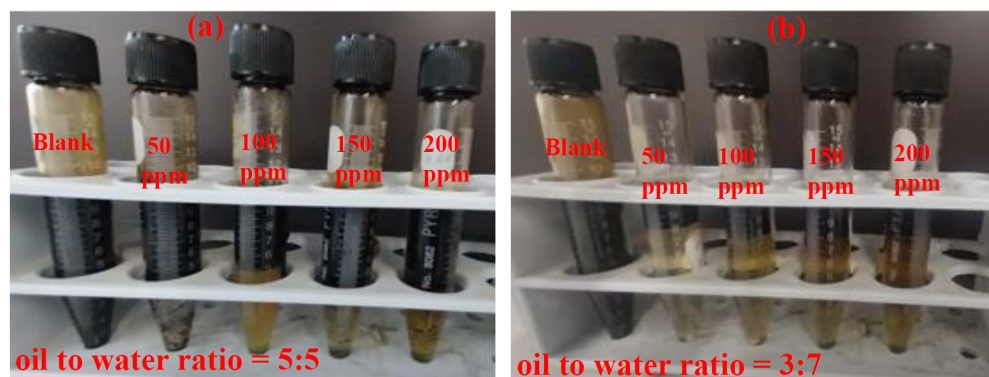


Figure 3—Oil-water separation driven by ND using the bottle method (a) oil to water is 5:5 (b) oil to water is 3:7.

Emulsion viscosity with and without ND

Bottle test technique has been a widely employed means to assess the performance of chemical demulsifiers to break crude oil emulsions. This method is very quick and simple but its reliability to really establish the mechanism and how demulsifiers destroy the emulsifying materials in stable emulsion is limited. As a result, rheology is another means of assessing the behavior of emulsion in the presence and absence of demulsifiers. Usually, in the bulk rheology measurement, emulsion viscosity is expected to diminish in the presence of a working chemical demulsifier by attacking the asphaltene molecules surrounding the emulsion droplets. Fig. 4 (a & b) shows the viscosity behavior of emulsions formed from oil to water ratios of 5:5 and 3:7 in the absence and presence of 100 ppm ND. Prior to the addition of ND, the viscosities of blank samples without ND were recorded to be 13.91 Pa.s and 58.09 Pa.s after 1800 seconds (30 minutes) for emulsion formulated from oil to water ratios of 5:5 and 3:7. Upon the addition of 100 ppm ND in both emulsions, a drastic decrease in the viscosities of both emulsions was noticed across the entire period of the viscosity check. Accordingly, the viscosity of emulsion formulated from oil to water ratios of 5:5 and 3:7 in the presence of 100 ppm ND after 1800 seconds was 0.24 Pa.s and 0.83 Pa.s, respectively. The reduction

in the viscosities of both emulsions in the presence of ND revealed that the emulsifying substances such as asphaltenes and resins surrounding the emulsion droplets were distorted (Mohammadian et al. 2018) systematically leading to the coalescence of emulsion droplets, formation of big globules and separation of oil and water into phases.

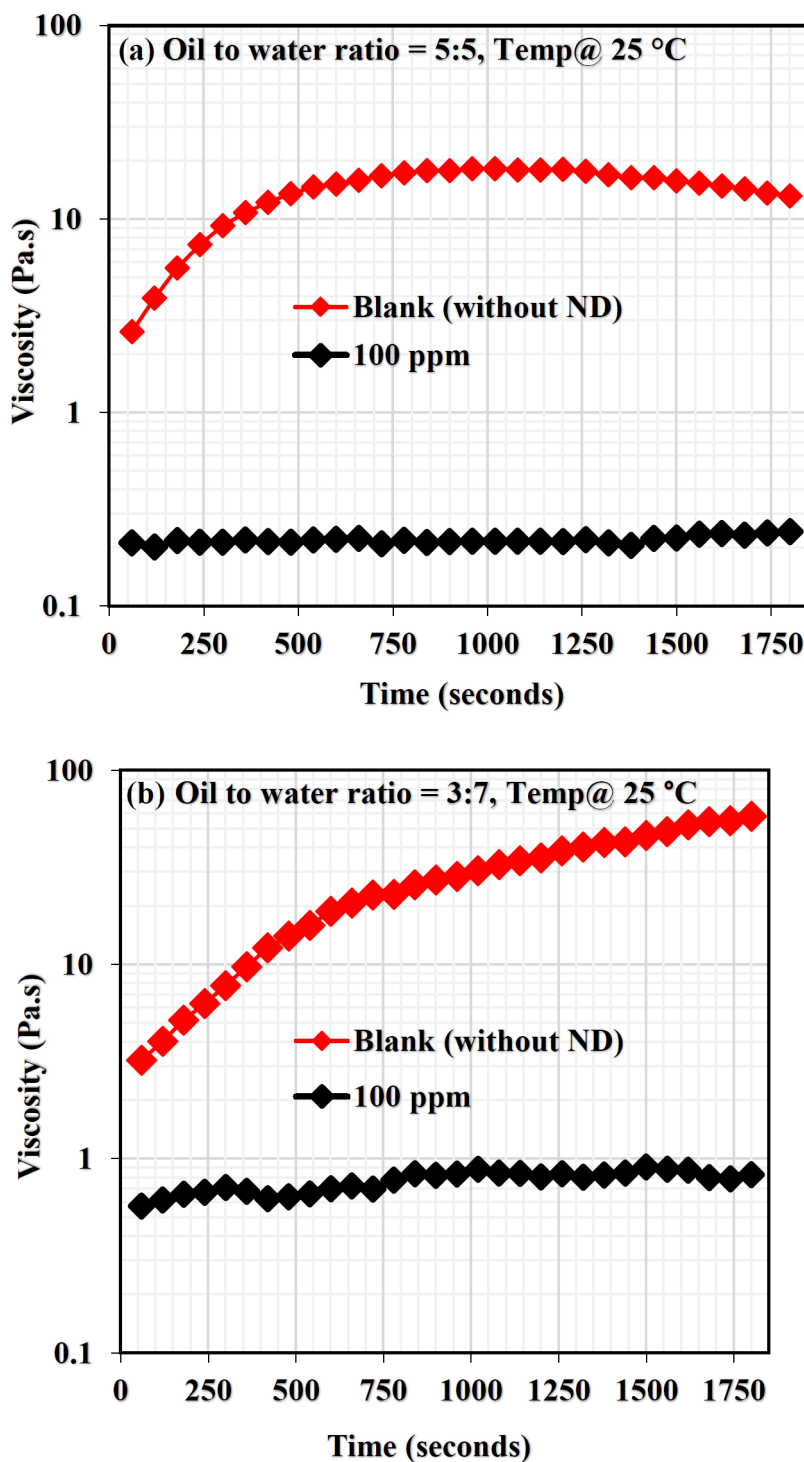


Figure 4—Emulsion viscosity behavior with and without ND (a) Oil to water ratio of 5:5 (b) Oil to water ratio of 3:7.

Emulsion stability via Turbiscan

Destabilization kinetics of the formed emulsions with and without ND was evaluated by Turbiscan. Using the Turbiscan aids the understanding and quantification of separation, sedimentation, flocculation and coalescence of fluids such as emulsion, colloidal solution, gel, foam, etc (Chen et al. 2023). As for the prepared crude oil emulsion, since the crude oil is dark in color, it would either obstruct or absorb the transmitted light, therefore, oil phase would exhibit a decrease in signal or no signal while water phase would push up the oil phase allowing more light signals to pass through. Hence, water phase would be indicated by an increasing amount of light been detected. The first scan is represented by red color, the last scan is purple color and the cell height from bottom to its top (i.e. from 0 to 50 mm). Shown in Fig. 5a is the destabilization kinetic profile of emulsion without ND at 70 °C. In the entire scanning process, it is obvious that light signals were obstructed or absorbed because the emulsion remained dark throughout the entire 24 hours of scanning and from the cell bottom to its top. On the other hand, Fig. 5b demonstrated that the light signals transmitted in the emulsion filled with 100 ppm ND. Evidently, light signals increased significantly to about 140% at the cell bottom (0 -3 mm) and the intensity of light signals diminished to about 100% at around 5 – 30 mm, indicating the separation of water and settling to the bottom while light signals were absorbed at 33 – 44 mm, suggesting the presence of crude oil in the region.

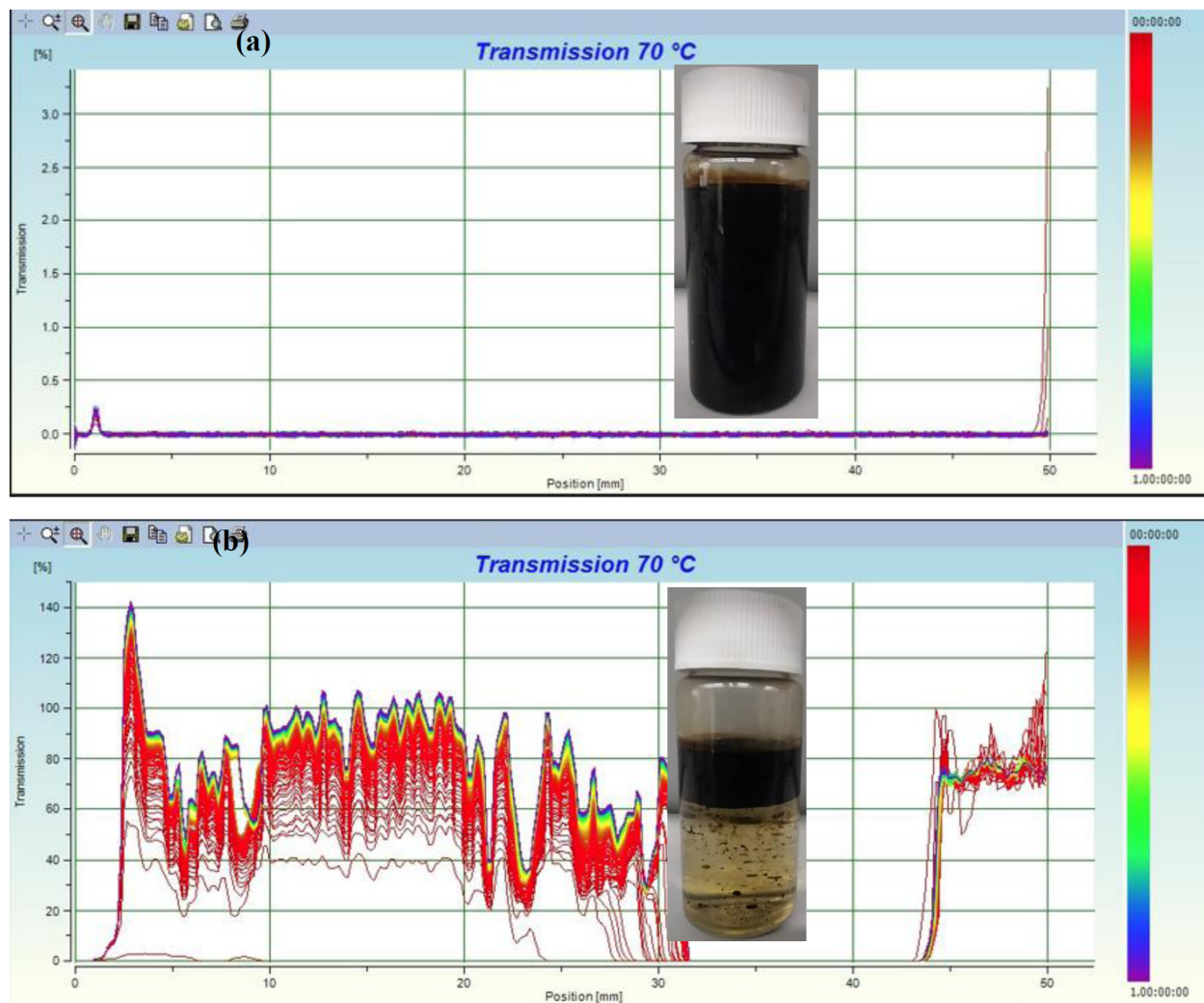


Figure 5—5 Light transmission profiles for (a) blank emulsion (b) emulsion filled with 100 ppm ND.

Conclusions

In this study, a new hydrocarbon based chemical demulsifier known as non-ionic demulsifier (ND) was developed and its effectiveness to break crude oil emulsions generated from varying oil to water ratios was conducted. The following conclusions were deduced:

1. Formulated emulsions using oil to water ratios of 5:5 and 3:7 were destabilized using ND concentrations of 50, 100, 150 and 200 ppm at demulsification temperature of 85 °C.
2. Various dosages of ND demulsified the formed emulsions with the highest %DE of 96% and 99.86% obtained at 100 ppm ND for oil to water ratios of 5:5 and 3:7, respectively.
3. Bulk viscosity measurements at 25 °C revealed the emulsion breaking mechanism in the presence and absence of ND. The addition of 100 ppm ND disrupted the stabilizing agents around the emulsion droplets causing emulsion viscosity to drop significantly as compared to the viscosity of emulsion without ND.
4. Turbiscan measurements evaluated at 70 °C described the destabilization kinetics of the produced emulsions before and after the incorporation of ND.

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Nomenclature

DE	demulsification efficiency, %
V	volume of water separated, mL
Vo	initial water volume, mL

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